

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I M Vishnu Vardhan Reddy (192572062) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Vishnu

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.

Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

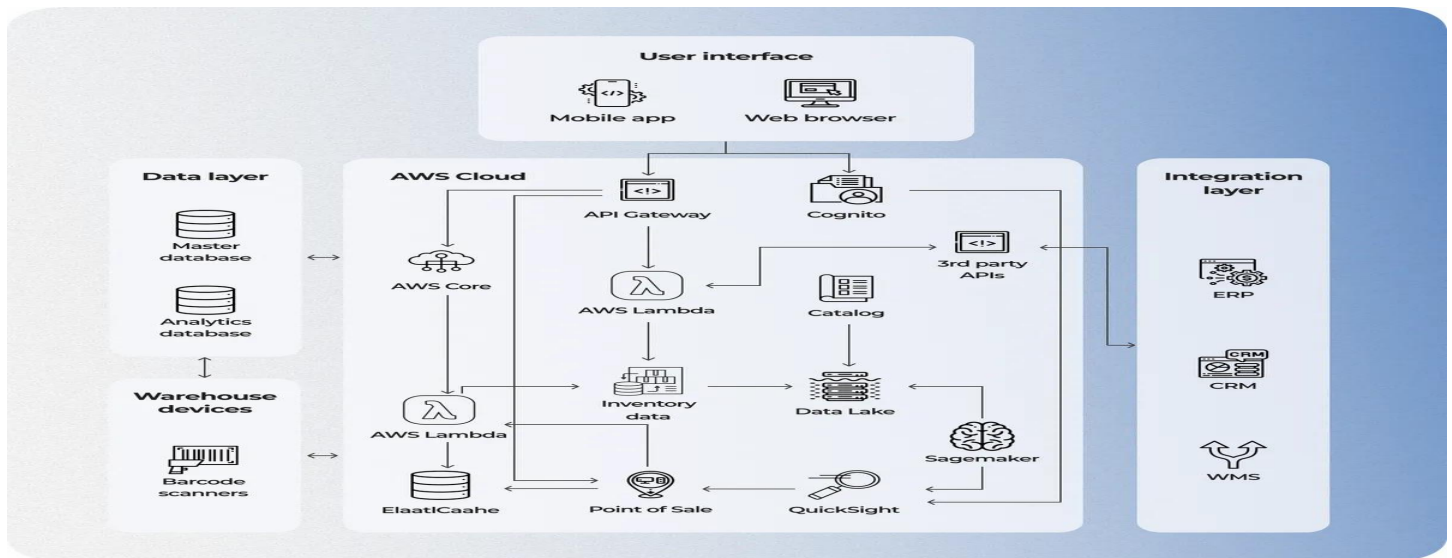
A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure

uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study 1: Smart Retail Inventory Management System (100 Marks)

1. Architecture Diagram (20 Marks)



Architecture Explanation

The Smart Retail Inventory Management System connects multiple retail stores to a centralized cloud platform.

Components

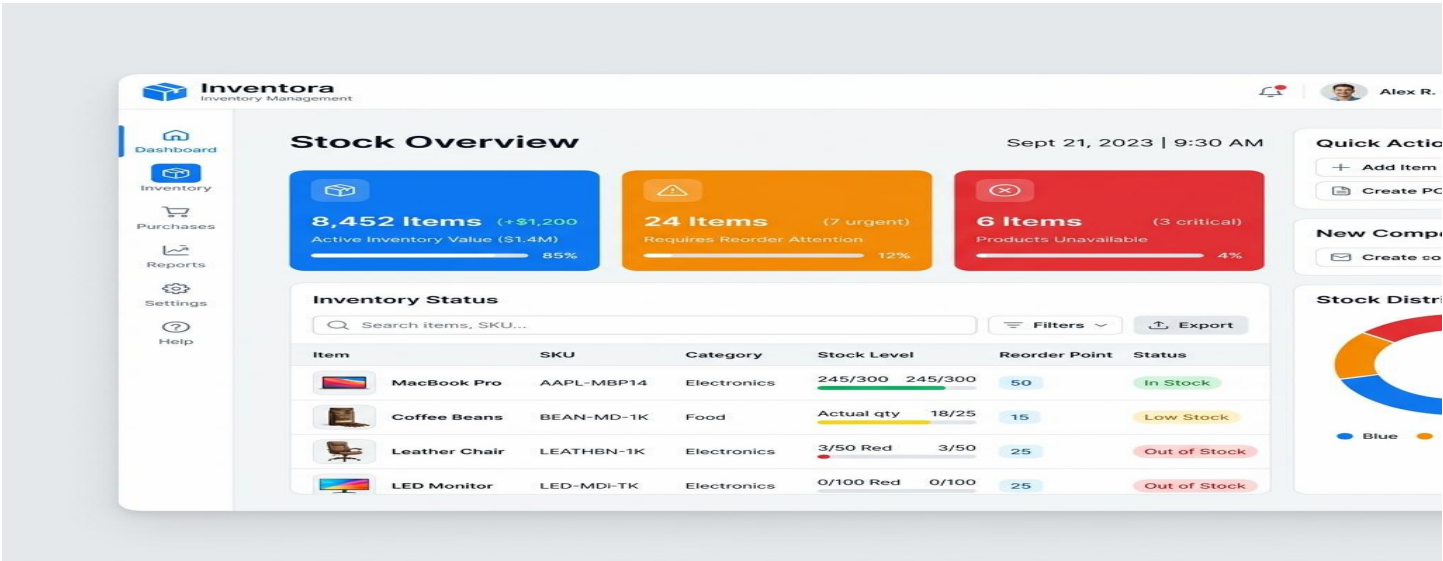
- POS Systems – Process customer purchases.
- Store Dashboard – Allows managers to monitor inventory.
- API Gateway – Receives all client requests securely.
- Cloud Services – Handles inventory, billing, and orders.
- Analytics Platform – Predicts future demand.
- Cloud Storage – Stores products, sales, and analytics.
- Security Layer – VPN, Firewall, HTTPS, Authentication.

Data Flow

1. Customer purchases through POS.
2. POS sends data via REST API.
3. API Gateway validates request.
4. Cloud updates inventory.
5. Data is stored in cloud database.
6. Analytics forecasts future stock demand.

2. Inventory Webpage (HTML/CSS/JavaScript) – 20 Marks

Output Preview



```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Smart Inventory</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<h1>Smart Retail Inventory</h1>
```

```
<div class="card">
```

```
<h2>Milk</h2>
```

```
<p id="milk">50</p>
```

```
<button onclick="updateStock('milk',1)">+</button>
```

```
<button onclick="updateStock('milk',-1)">-</button>
```

```
</div>
```

```
<div class="card">
```

```
<h2>Bread</h2>
```

```
<p id="bread">30</p>
```

```
<button onclick="updateStock('bread',1)">+</button>
```

```
<button onclick="updateStock('bread',-1)">-</button>
```

```
</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

Features

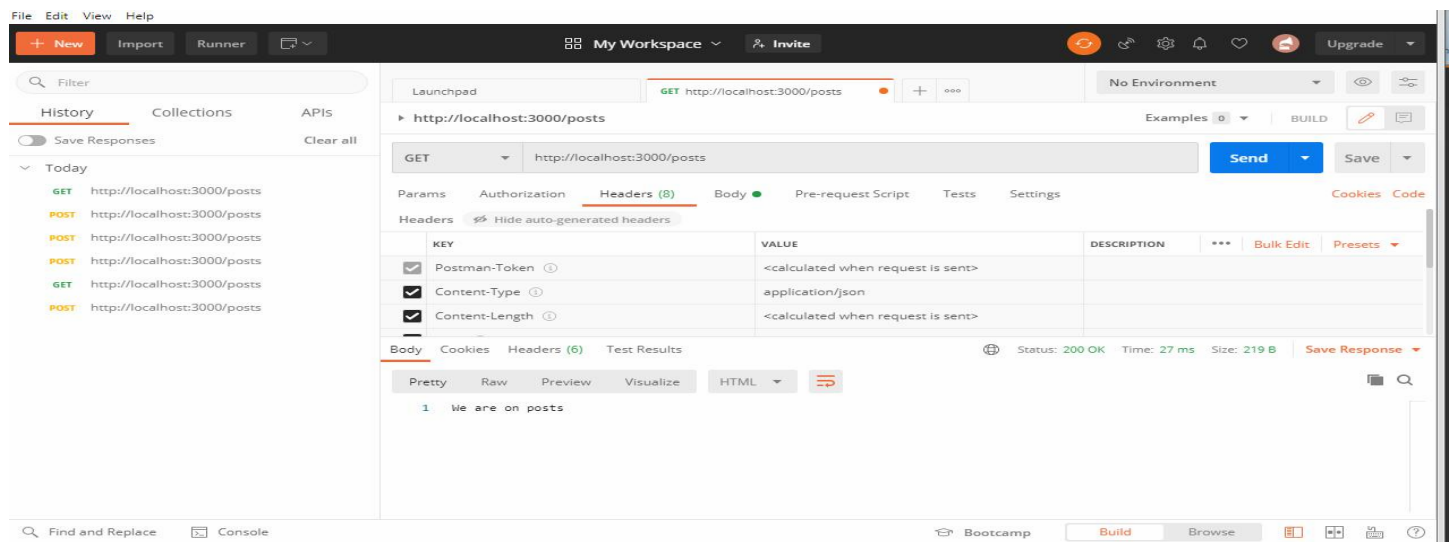
- Product display
- Live stock update
- Simple dashboard
- Responsive layout

3. REST API Testing using Postman (20 Marks)

Use JSONPlaceholder for demonstration.

Base URL

<https://jsonplaceholder.typicode.com>



4. Report with Screenshots and Explanation (20 Marks)

Report Structure

1. Title
2. Objective
3. Introduction
4. Architecture Diagram
5. Components Explanation
6. Inventory Webpage
7. API Testing Screenshots
8. Advantages
9. Conclusion

Screenshot Placeholders

- Architecture Diagram
- Inventory Webpage
- GET Request
- POST Request
- PUT Request
- DELETE Request

5. Demo Presentation (20 Marks)

PPT Outline (10 Slides)

Slide Content

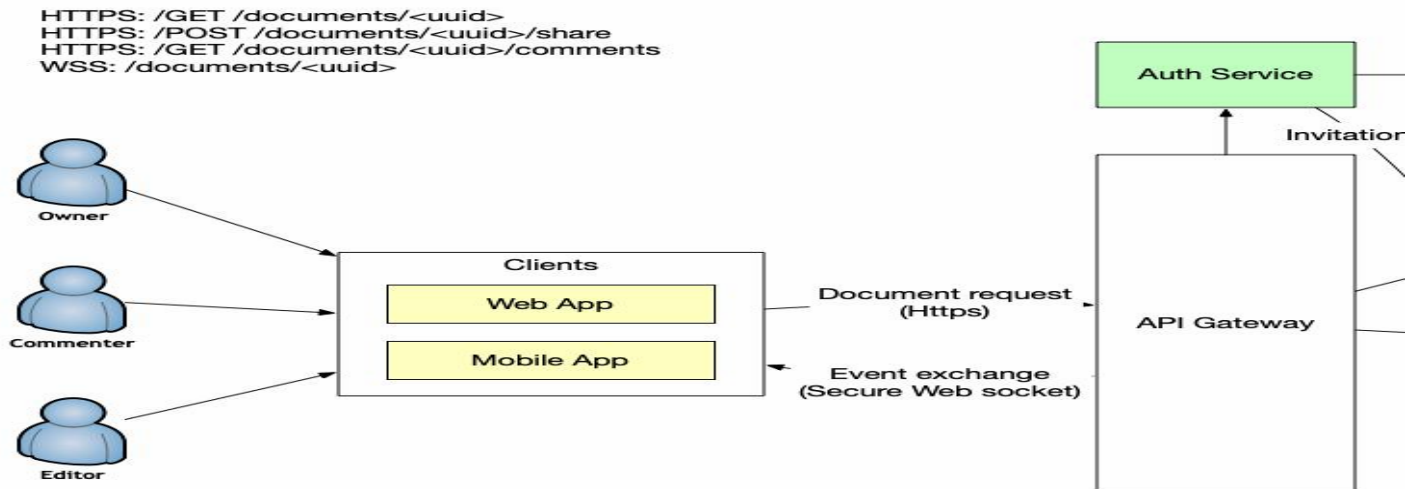
- 1 Title
- 2 Problem
- 3 Architecture
- 4 Components
- 5 Inventory Webpage
- 6 API Testing
- 7 Analytics
- 8 Security

9 Advantages

10 Conclusion

Case Study 2: Cloud-Based Document Collaboration System (100 Marks)

1. Architecture Diagram (20 Marks)



Components

- Browser Clients
- Mobile Devices
- Web APIs
- Collaboration Server
- Version Control
- Distributed Storage
- Authentication
- Encryption

2. Component Identification (10 Marks)

Component	Function
Browser	User access
Web API	Synchronization
Collaboration Server	Real-time editing

Version Control History tracking

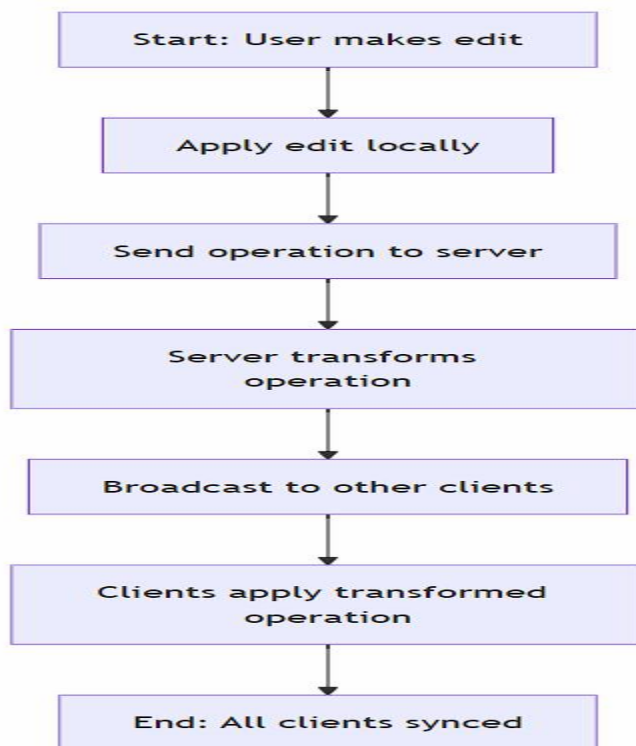
Cloud Storage Document storage

Identity Management Login

Encryption Security

3. Workflow Explanation (20 Marks)

Real-Time Editing Process



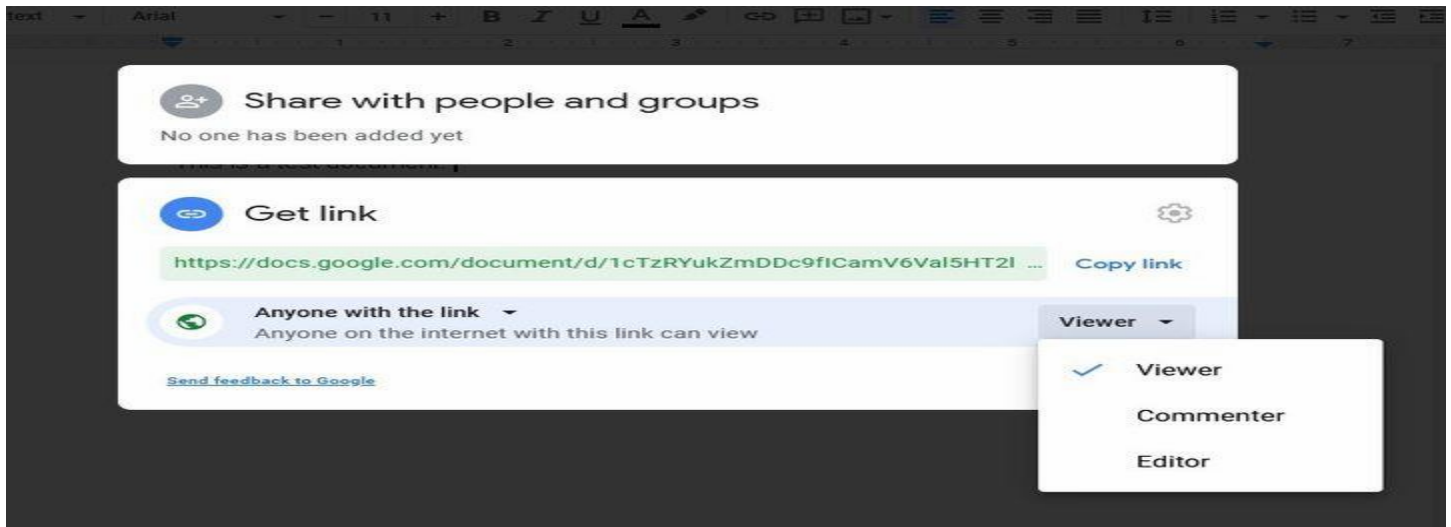
Workflow

1. User opens document.
2. Browser requests API.
3. Server loads latest version.
4. User edits.
5. Changes sync instantly.
6. Other users receive updates.
7. Version history is saved.

Benefits

- Low latency
- High availability
- Automatic synchronization

4. Google Docs Practical (20 Marks)



Steps

1. Open Google Docs.
2. Create a document.
3. Add text.
4. Click Share.
5. Enter another email.
6. Give Editor permission.
7. Edit simultaneously.

Observe

- Live cursor movement
- Instant synchronization
- Version History

5. Cloud Storage Demonstration (20 Marks)

Use Google Drive.

Steps

1. Upload document.

2. Open from another device.
3. Edit.
4. Check Version History.

Features

- Backup
- Synchronization
- Multiple devices
- Recovery

6. Security Analysis (10 Marks)

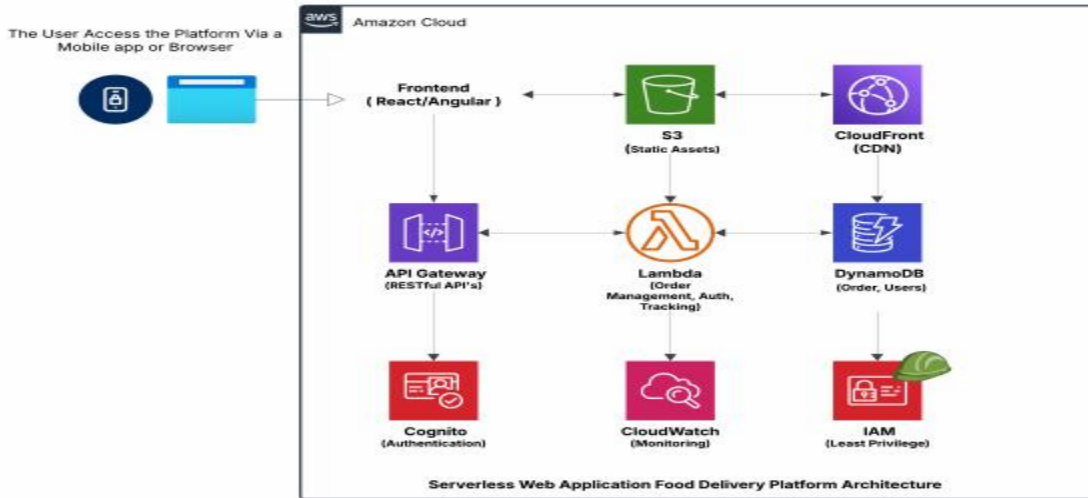
Security	Purpose
HTTPS	Encrypted communication
AES Encryption	Protect documents
OAuth	Secure login
Access Control	Permissions
Version Control	Recovery

Advantages

- Secure collaboration
- Global availability
- Automatic backup
- Real-time editing

Case Study 3: Online Food Ordering Platform (100 Marks)

1. Architecture Diagram (20 Marks)

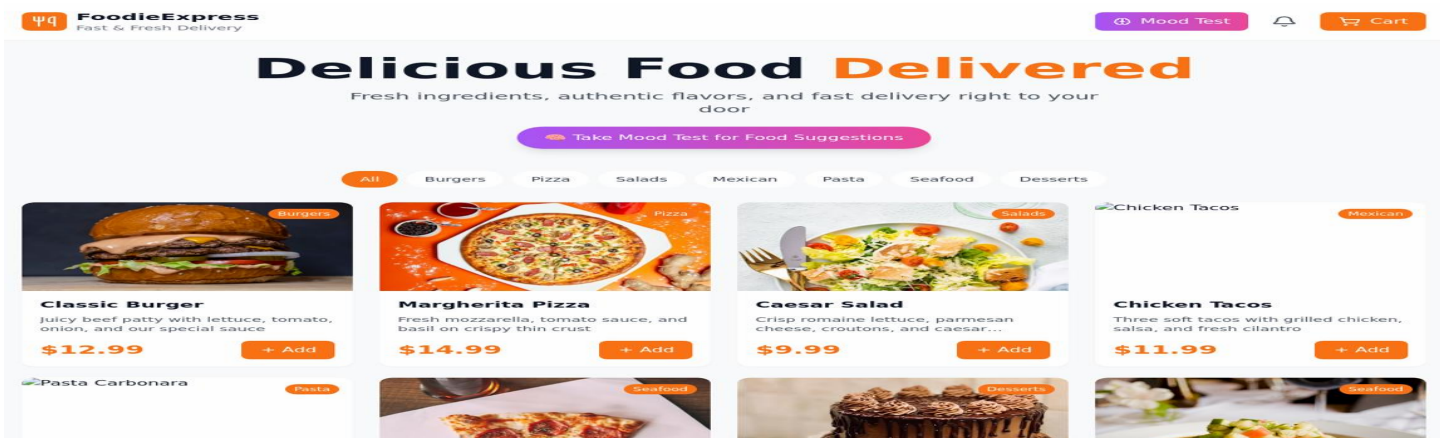


Components

- Mobile App
- Web Browser
- REST API
- Kubernetes
- Cloud Storage
- Database
- Authentication
- Delivery Tracking

2. Online Food Ordering Webpage (HTML/CSS/JS) – 20 Marks

Output Preview



```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Food Ordering</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<h1>Cloud Food Ordering</h1>
```

```
<div class="food">
```

```
<h2>Pizza</h2>
```

```
<p>₹299</p>
```

```
<button onclick="addCart('Pizza')">Add</button>
```

```
</div>
```

```
<div class="food">
```

```
<h2>Burger</h2>
```

```
<p>₹149</p>
```

```
<button onclick="addCart('Burger')">Add</button>
```

```
</div>
```

```
<h2>Cart</h2>
```

```
<ul id="cart"></ul>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
• </html>
```

Menu display

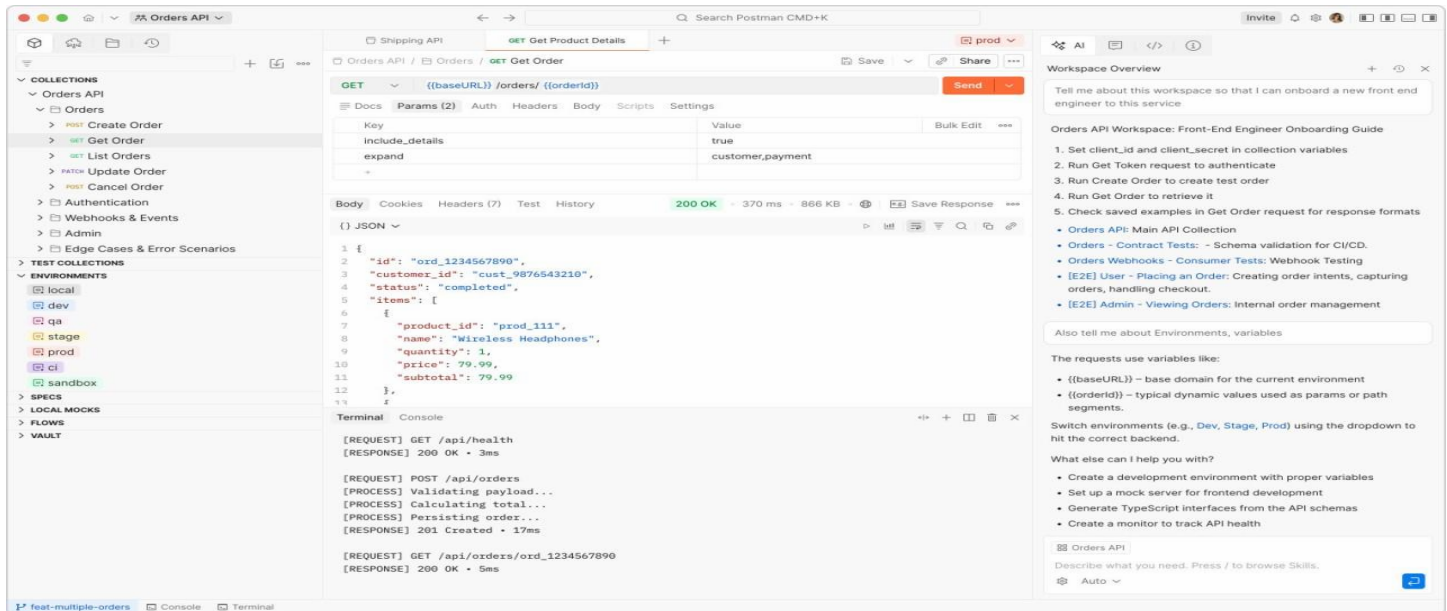
- Add to Cart
- Responsive interface

3. REST API Testing using Postman (20 Marks)

Base URL

<https://jsonplaceholder.typicode.com>

GET Menu



POST Order

Body

```
{  "food": "Pizza",  "quantity": 2}
```

PUT Order

```
{  "quantity": 3}
```

DELETE Order

Method

DELETE

Expected Status

200 OK

4. Report with Screenshots (20 Marks)

Report Contents

1. Introduction
2. Architecture
3. Components
4. Food Ordering Webpage
5. API Testing
6. Security
7. Advantages
8. Conclusion

Screenshot List

- Homepage
- Add to Cart
- GET API
- POST API
- PUT API
- DELETE API

5. Demo Presentation (20 Marks)

PPT Structure

Slide Content

- 1 Title
- 2 Problem
- 3 Architecture
- 4 Webpage

- 5 REST API
- 6 Cloud Storage
- 7 Kubernetes
- 8 Security
- 9 Benefits
- 10 Conclusion

Demo Flow (5 Minutes)

- Open Food Ordering webpage.
- Add Pizza and Burger to cart.
- Show cart update.
- Open Postman.
- Perform GET, POST, PUT, DELETE requests.
- Explain Kubernetes scaling and cloud storage.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
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Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized